

Table 4: Hiring, Separations, and Promotions

	Hiring	Separation	Promotion (rate)
Treat $\times$ Post	−4.006*** (0.221)	−1.089*** (0.163)	0.033** (0.014)
Pre-period mean	5.059	4.154	0.822
Observations	7,941,179	7,941,179	7,690,581
Clusters (firms)		280,866	

Notes: This table reports the estimated β from Equation 15. Estimation begins in 2018 to exclude early-sample trends. Standard errors clustered by firm in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6 Conclusion

This paper provides early, large-scale evidence that the diffusion of GenAI since 2023 is associated with *seniority-biased* employment effects within firms. Using résumé and job-posting data linked to roughly 281,000 U.S. firms, together with a firm-level adoption measure based on “GenAI integrator” vacancies, we document a decline in junior employment relative to senior employment at adopting firms compared with non-adopting firms. This decline is concentrated in high-exposure occupations: within adopting firms, junior employment in high-exposure occupations contracts relative to low-exposure occupations after 2022, while the corresponding senior series continues to rise. Task-level evidence from job postings points to the same mechanism: GenAI-exposed tasks become less likely to appear in junior task bundles at adopting firms. Finally, decomposing employment flows shows that the decline is driven primarily by slower hiring rather than increased separations.

These findings should be interpreted with caution. Adopting firms differ systematically in size, workforce composition, and industry. Although our research designs absorb many observable and unobservable differences across firms, sectors, and time, unobserved confounding factors may remain. We therefore examine four prominent alternative explanations: monetary tightening, post-pandemic hiring dynamics, the expansion of remote work, and labor-supply shocks. These tests support our interpretation, but they cannot rule out all alternative explanations. In addition, our adoption measure captures deliberate organizational integration of GenAI but may miss informal or “silent” adop-

tion within firms. Finally, because GenAI is still in the early stages of diffusion, longer-run adjustments in training, task allocation, wages, and internal career ladders may either attenuate or amplify the effects documented here.

Even with these caveats, the results suggest that GenAI adoption may be shifting labor demand away from entry-level tasks and narrowing the lower rungs of internal career ladders. Because early-career jobs are central to skill formation and lifetime earnings growth, such changes may have consequences beyond short-run employment. Whether these patterns persist, and whether firms respond by redesigning training, tasks, or career progression, remains an important question for future research.

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A.3 Heterogeneity by Educational Background

This appendix examines whether the decline in junior employment varies by workers' educational background, as referenced in Section 5.2.3. We first describe the construction of the school-quality variable, then present the heterogeneity results.

A.3.1 Construction of the School-Quality Variable

To capture educational background, we construct a position-level measure of school quality. For each position, we assign the institution where the worker most recently studied, provided that the education ended no later than one year after the job start. If no such record exists, we use the first recorded institution, provided it began before the job start date. Each institution is then assigned a GPT-4-based quality rating on a five-point scale, produced using OpenAI's GPT-4o-mini model (see Supplemental Appendix A.11.2 for the full prompt). A score of 1 corresponds to Ivy League and other globally elite institutions; 2 to highly respected international universities; 3 to strong national or regional schools; 4 to lower-tier but standard institutions; and 5 to minimally selective or non-accredited institutions.

A.3.2 Results

We re-estimate Equation 34 for junior employment separately within each of the five school-quality tiers. For clarity of presentation, we use a single post-treatment indicator for quarters after 2023Q1 instead of a full set of time dummies.

The results, shown in Figure A.12, reveal a U-shaped pattern. Juniors from tier-3 universities experienced the steepest relative declines in employment, while those from tiers 2 and 4 also saw significant reductions. Juniors from tier-1 and tier-5 universities experienced the smallest decline.³³ Figure A.13 reports the time-series version of these estimates, with a full set of time dummies in place of the single post-2023 indicator.

To help interpret this pattern, Figures A.14 and A.15 show average predicted salaries³⁴

³³In the analysis, we exclude positions where the worker's most recent educational entry is reported as a high school (approximately 5 percent of positions). Including these positions, based on high school quality, has virtually no effect on the results.

³⁴Actual position-level salaries are unavailable. However, Revelio Labs provides a predicted salary

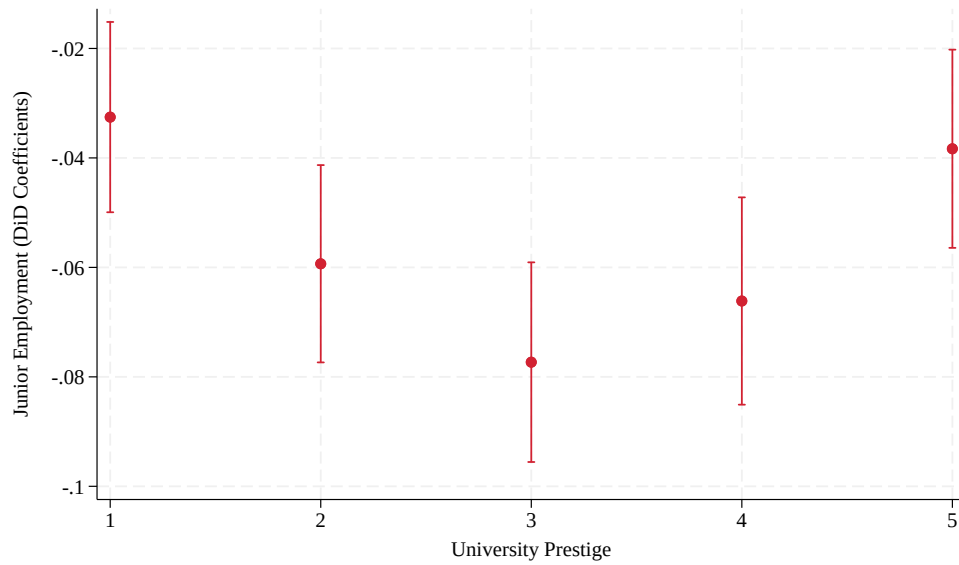


Figure A.12: Heterogeneity by School Quality

Notes: Each bar reports the estimated coefficient from re-estimating Equation 34 for junior employment separately within each of the five school-quality tiers. The specification replaces time dummies with a single post-treatment indicator for quarters after 2023Q1. Estimation begins in 2018 to exclude early-sample trends. Standard errors are clustered at the firm level.

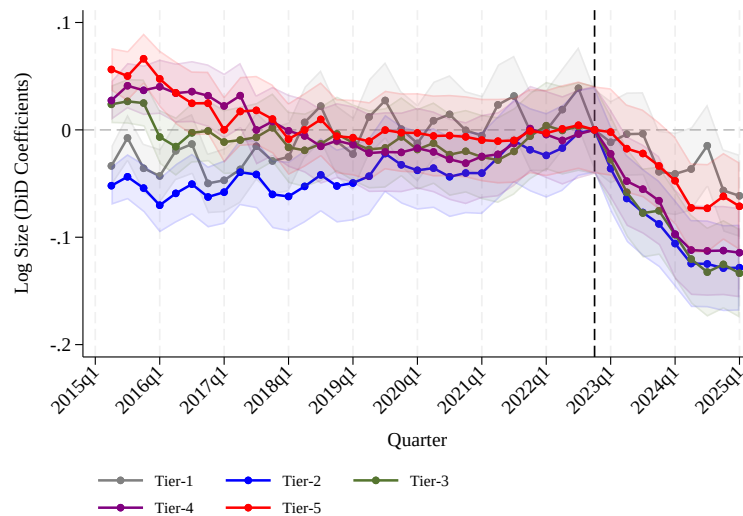


Figure A.13: DiD by Educational Background—Time Series (Juniors)

Notes: This figure plots β_j from Equation 34, estimated separately for each of the five school-quality tiers. Standard errors are clustered at the firm level.

variable imputed using position-specific information. For more details, see: <https://www.data-dictionary.reveliolabs.com/methodology.html#salary>.